

INFRASTRUKTUR SERVER INTERNAL

Otomatisasi CI/CD & Lingkungan Review Dinamis

Implementasi self-hosted GitLab, Traefik, dan manajemen konfigurasi terpusat.

Arsitektur Ekosistem DevOps



GitLab CI/CD

Pusat eksekusi pipeline.



Code-Server

Injeksi konfigurasi terpusat.



Harbor Registry

Distribusi custom image.



Portainer

Monitoring dan orkestrasi node.



Traefik Proxy

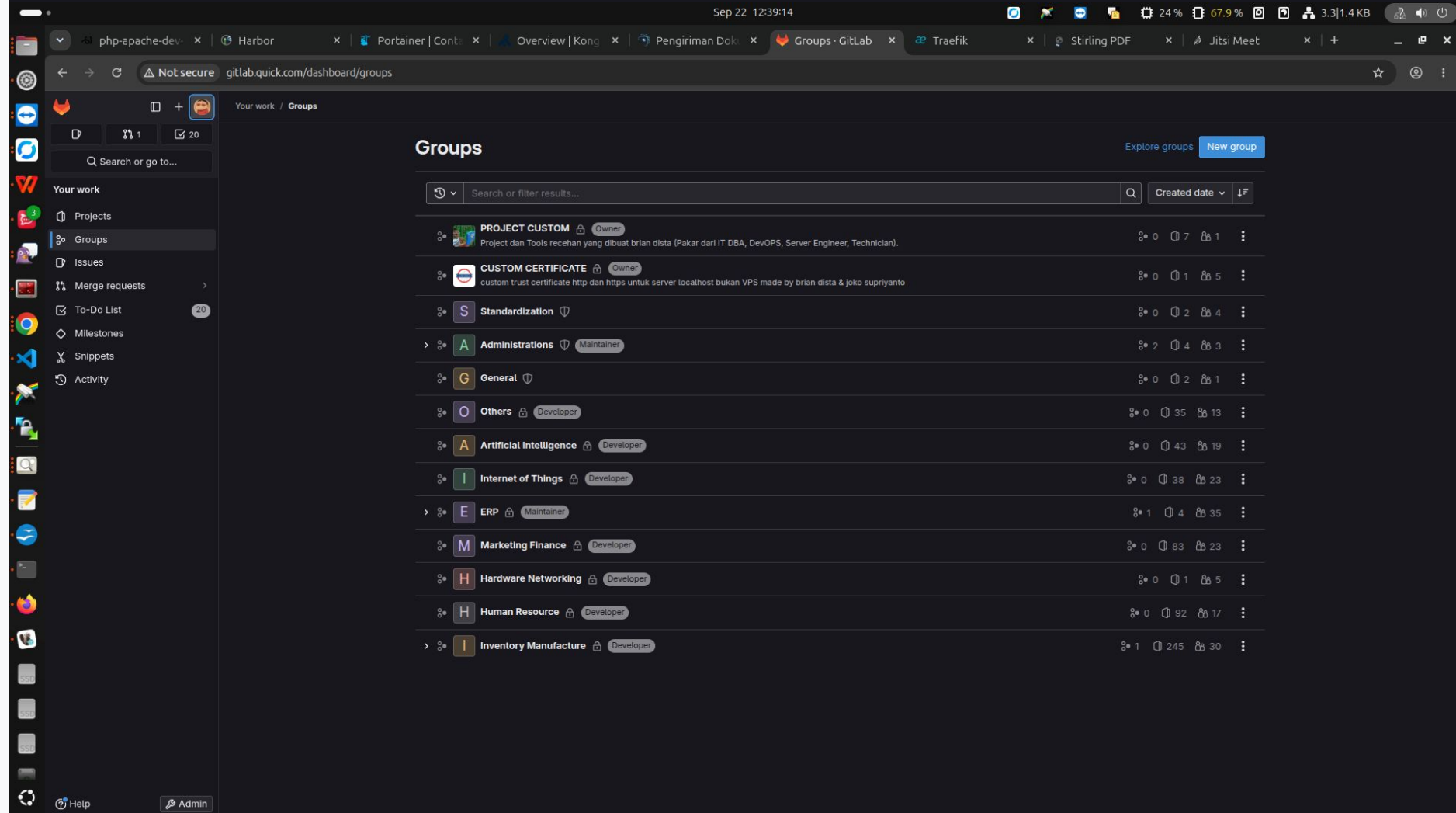
Manajemen routing dinamis.

MANAJEMEN KODE

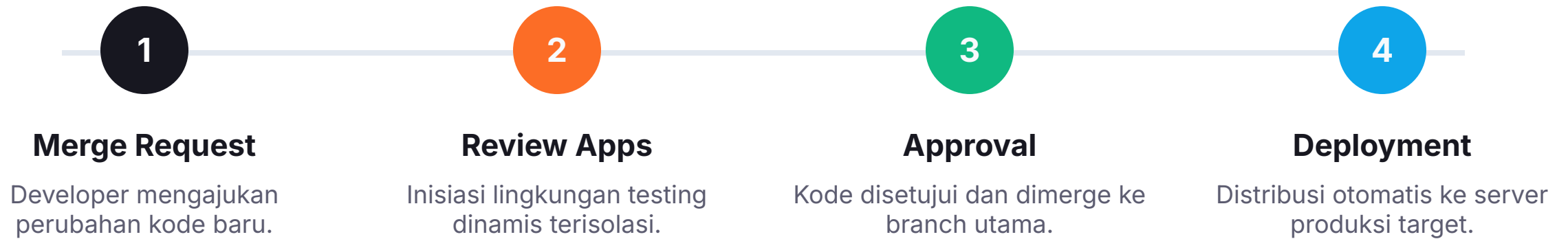
Struktur Repositori

Klasifikasi hierarkis berdasarkan grup divisi dan nama proyek.

- ✓ Akses terkontrol.
- ✓ Rapih dan terukur.



Alur Kerja Otomatisasi



```
1 services:
2   web:
3     image: reg.docker.quick.com/webdevops/php-apache-dev:5.6-0.0.1
4     container_name: $CI_PROJECT_ROOT_NAMESPACE.$CI_PROJECT_NAME.$CI_MERGE_REQUEST_ID.web
5     restart: unless-stopped
6     user: "$USER_GROUP_ID"
7     userns_mode: "host"
8     environment:
9       - WEB_PHP_TIMEOUT=5000
10      # - XDEBUG_MODE=profile,trace
11      # - XDEBUG_START_WITH_REQUEST=yes
12      # - XDEBUG_PROFILER_ENABLE=1
13      # - XDEBUG_PROFILER_OUTPUT_DIR=/xdebug
14      - CI_ENV=development
15      - php.session.save_path=/tmp/php-sessions
16     volumes:
17       - ./www:/app
18       - ./www/xdebug:/xdebug
19       - ./secrets:/secrets
20
21     command: >
22       sh -c "
23         curl -sS https://getcomposer.org/installer | php -- --2 --install-dir=/usr/local/bin --filename=composer &&
24         apache2-foreground
25       "
26
27     deploy:
28       resources:
29         limits:
30           memory: 1000M
31
32     labels:
33       - "traefik.enable=true"
34       - "traefik.http.routers.$CI_MERGE_REQUEST_ID-$CI_COMMIT_SHA.rule=Host('$CI_MERGE_REQUEST_ID.$CI_PROJECT_NAME.$CI_PROJECT_ROOT_NAMESPACE.staging.quick.com')"
35       - "traefik.http.routers.$CI_MERGE_REQUEST_ID-$CI_COMMIT_SHA.entrypoints=http"
36       - "traefik.http.routers.$CI_MERGE_REQUEST_ID-$CI_COMMIT_SHA.tls=false"
37
38     networks:
39       - traefik
40
41 networks:
42   traefik:
43     name: traefik proxy
44     external: true
```

INJEKSI DEPENDENSI

Konfigurasi Terpusat

Pengambilan template Docker Compose dan konfigurasi koneksi on-the-fly.

Bebas hardcode.

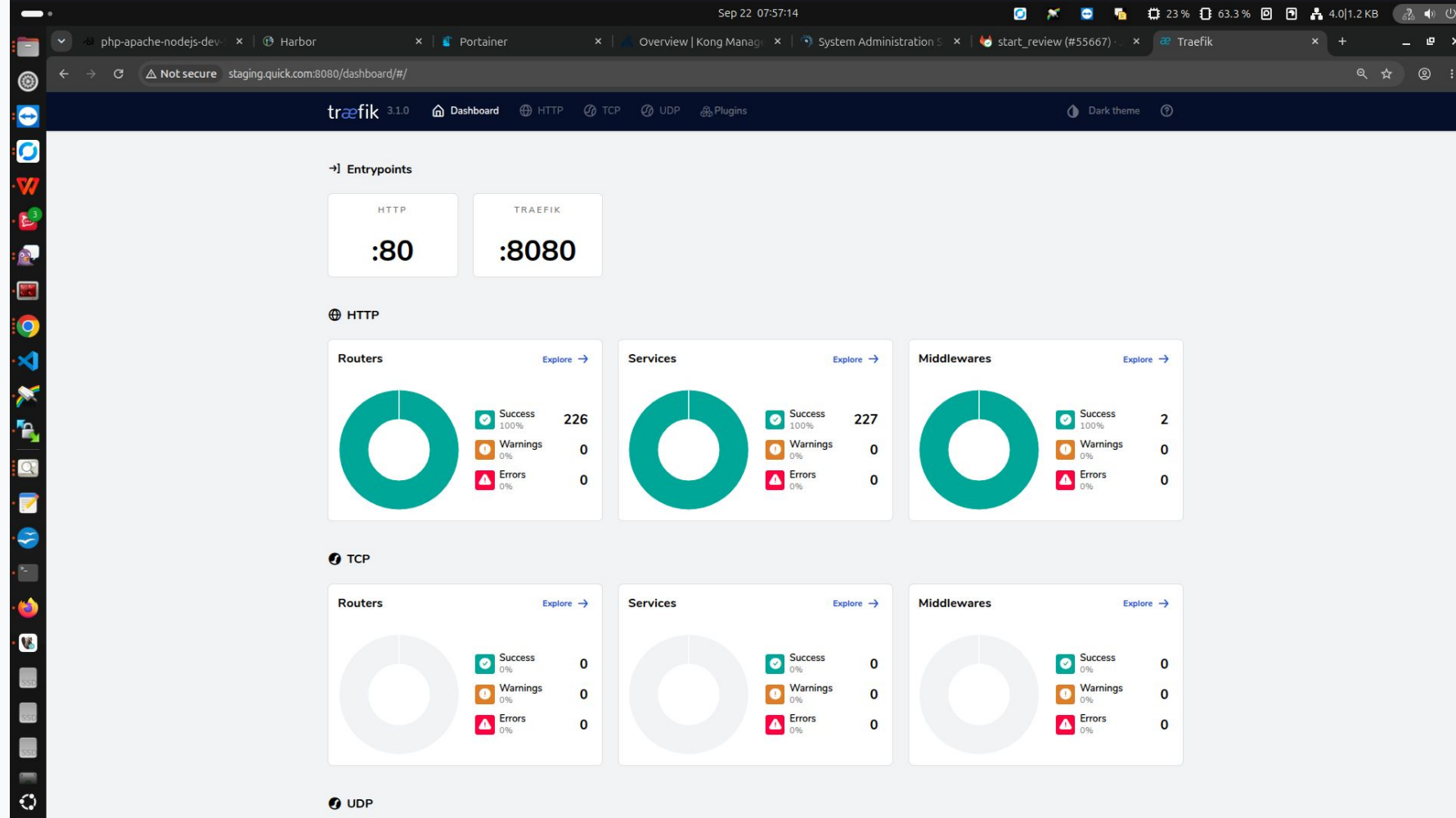
Setiap proyek mengambil file spesifik jalurnya via perintah curl otomatis.

NETWORK MANAGEMENT

Routing Dinamis Traefik

Mengelola puluhan lingkungan Review Apps secara paralel.

- 🛡️ Resolusi DNS otomatis.
- 🛡️ Anti tabrakan port antar kontainer.



Validasi Kode Instan

Walkthrough: Setup VS C

Harbor

Portainer

Viewing route | Kong Ma

System Administration S

start_review (#55667)

Not secure

gitlab.quick.com/erp/web/-/jobs/55667

ERP / Web / Jobs / #55667

1

20

Search or go to...

Project

Web

Pinned

Issues

Merge requests

Pipeline editor

Pipelines

Branches

Manage

Plan

Code

Build

Pipelines

Pipeline editor

Pipeline schedules

Artifacts

Secure

Deploy

Operate

Monitor

Analyze

Settings

Help

Admin

Search visible log output

🔍

📄

↶

↷

⬆

⬇

⬇

⬆

44

- Installing setasign/fpdf (v2.6.0): Downloading (connecting...)

Downloading (0%)

Downloading (1)

45

- Installing setasign/fpdf (1.8.6): Downloading (connecting...)

Downloading (0%)

Downloading (5%)

Do

46

- Installing clegginabox/pdf-merger (dev-master 7383e41): Cloning 7383e41bfc

Downloading (0%)

Downloading (45%)

Down

47

- Installing psr/cache (1.0.1): Downloading (connecting...)

Downloading (0%)

Downloading (35%)

Down

48

- Installing firebase/php-jwt (v5.5.1): Downloading (connecting...)

Downloading (0%)

Downloading (80%)

Down

49

- Installing google/auth (v1.19.1): Downloading (connecting...)

Downloading (0%)

Downloading (5%)

Do

50

- Installing mykyawhtun/pdfmerger (dev-master a0e6fca): Cloning a0e6fcaf10

Downloading (0%)

Downloading (45%)

Down

51

- Installing ramsey/uuid (3.x-dev ad596eb): Cloning ad596ebba

Downloading (0%)

Downloading (75%)

Down

52

- Installing sentry/sentry (1.11.0): Downloading (connecting...)

Downloading (0%)

Downloading (100%)

Down

53

- Installing vlucas/phpdotenv (4.3.x-dev 67a491d): Cloning 67a491df68

Downloading (0%)

Downloading (50%)

Down

54

firebase/php-jwt suggests installing paragonie/sodium_compat (Support EdDSA (Ed25519) signatures when libsodium is not present)

Downloading (0%)

Downloading (55%)

Down

55

google/auth suggests installing phpseclib/phpseclib (May be used in place of OpenSSL for signing strings or for token management. Please require version ^2.)

Downloading (0%)

Downloading (100%)

Down

56

ramsey/uuid suggests installing ext-libsodium (Provides the PECL libsodium extension for use with the SodiumRandomGenerator)

Downloading (0%)

Downloading (100%)

Down

57

ramsey/uuid suggests installing ext-uuid (Provides the PECL UUID extension for use with the PeclUuidTimeGenerator and PeclUuidRandomGenerator)

Downloading (0%)

Downloading (100%)

Down

58

ramsey/uuid suggests installing moontoast/math (Provides support for converting UUID to 128-bit integer (in string form).)

Downloading (0%)

Downloading (100%)

Down

59

ramsey/uuid suggests installing paragonie/random-lib (Provides RandomLib for use with the RandomLibAdapter)

Downloading (0%)

Downloading (100%)

Down

60

ramsey/uuid suggests installing ramsey/uuid-console (A console application for generating UUIDs with ramsey/uuid)

Downloading (0%)

Downloading (100%)

Down

61

ramsey/uuid suggests installing ramsey/uuid-doctrine (Allows the use of Ramsey\Uuid\Uuid as Doctrine field type.)

Downloading (0%)

Downloading (100%)

Down

62

sentry/sentry suggests installing monolog/monolog (Automatically capture Monolog events as breadcrumbs)

Downloading (0%)

Downloading (100%)

Down

63

Generating autoload files

Downloading (0%)

Downloading (100%)

Down

64

8 packages you are using are looking for funding.

Downloading (0%)

Downloading (100%)

Down

65

Use the 'composer fund' command to find out more!

Downloading (0%)

Downloading (100%)

Down

66

\$ if [-f compose.yml]; then # collapsed multi-line command

Downloading (0%)

Downloading (100%)

Down

67

% Total % Received % Xferd Average Speed Time Time Time Current

Downloading (0%)

Downloading (100%)

Down

68

Dload Upload Total Spent Left Speed

Downloading (0%)

Downloading (100%)

Down

69

100 73685 100 73685 0 0 3182k 0 --:--:-- --:--:-- --:--:-- 3270k

Downloading (0%)

Downloading (100%)

Down

70

\$ if [-f compose.yml]; then # collapsed multi-line command

Downloading (0%)

Downloading (100%)

Down

71

% Total % Received % Xferd Average Speed Time Time Time Current

Downloading (0%)

Downloading (100%)

Down

72

Dload Upload Total Spent Left Speed

Downloading (0%)

Downloading (100%)

Down

73

100 43 100 43 0 0 6715 0 --:--:-- --:--:-- --:--:-- 7166

Downloading (0%)

Downloading (100%)

Down

74

\$ if [-f compose.yml]; then # collapsed multi-line command

Downloading (0%)

Downloading (100%)

Down

75

php-fpm:php-fpm: stopped

Downloading (0%)

Downloading (100%)

Down

76

php-fpm:php-fpm: started

Downloading (0%)

Downloading (100%)

Down

77

Job succeeded

Downloading (0%)

Downloading (100%)

Down

Duration: 3 minutes 24 seconds

Finished: 27 minutes ago

Queued: 1 second

Timeout: 1h (from project)

Runner: #15 (jVhQYZxQE) GitLab shell runner in staging server

Tags: staging-shell-runner

Commit 46c69ed0 in 16000

fix: Input aset opname image

Pipeline #24230 Blocked for 16000

with bugfix/T0828/of-i-asset-opname

review_apps

Related jobs

stop_review

start_review

Implementation By BRIAN DISTA

Custom Docker Image



Bypass limitasi image publik.



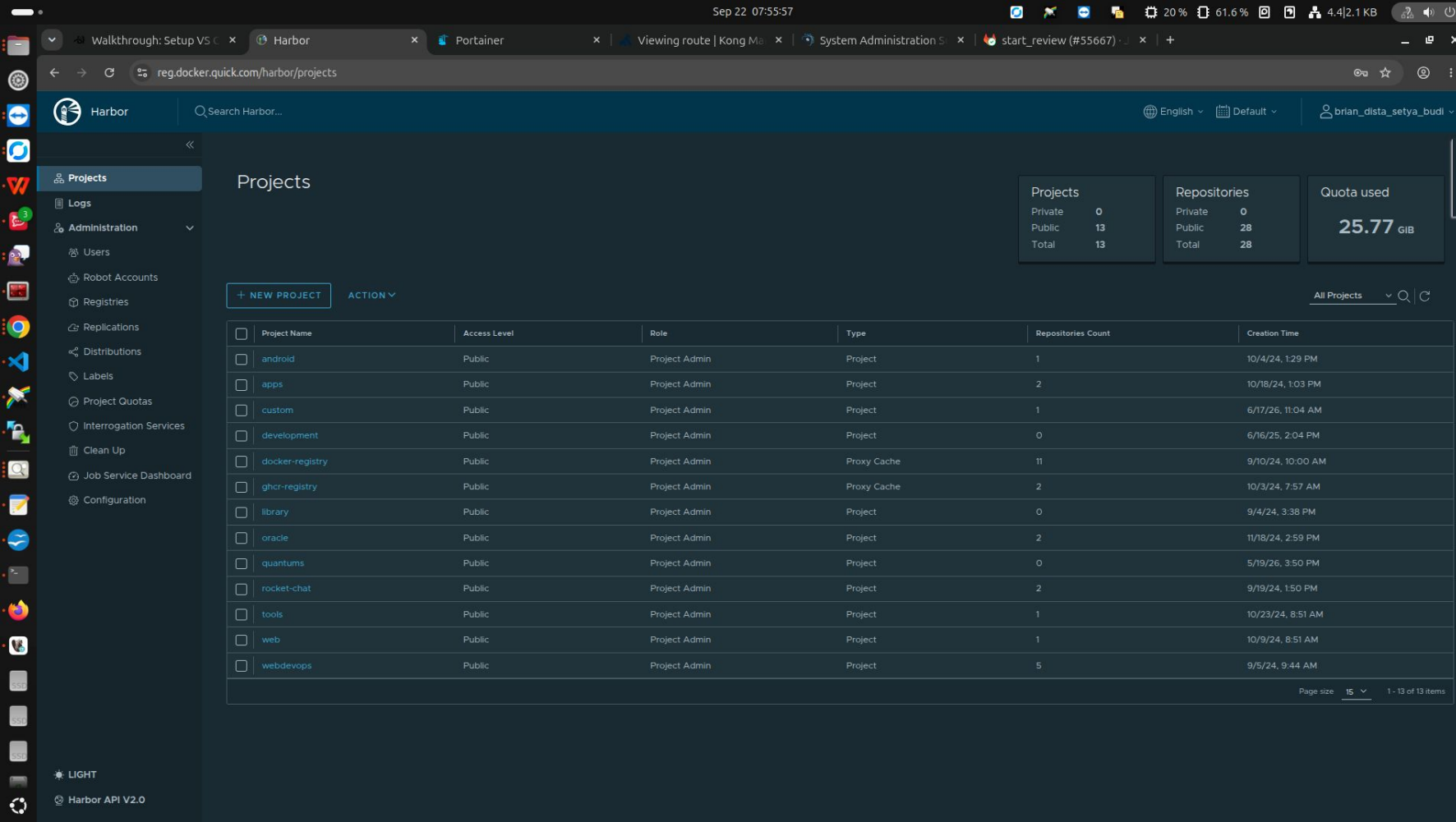


IMAGE DISTRIBUTION

Private Harbor Registry

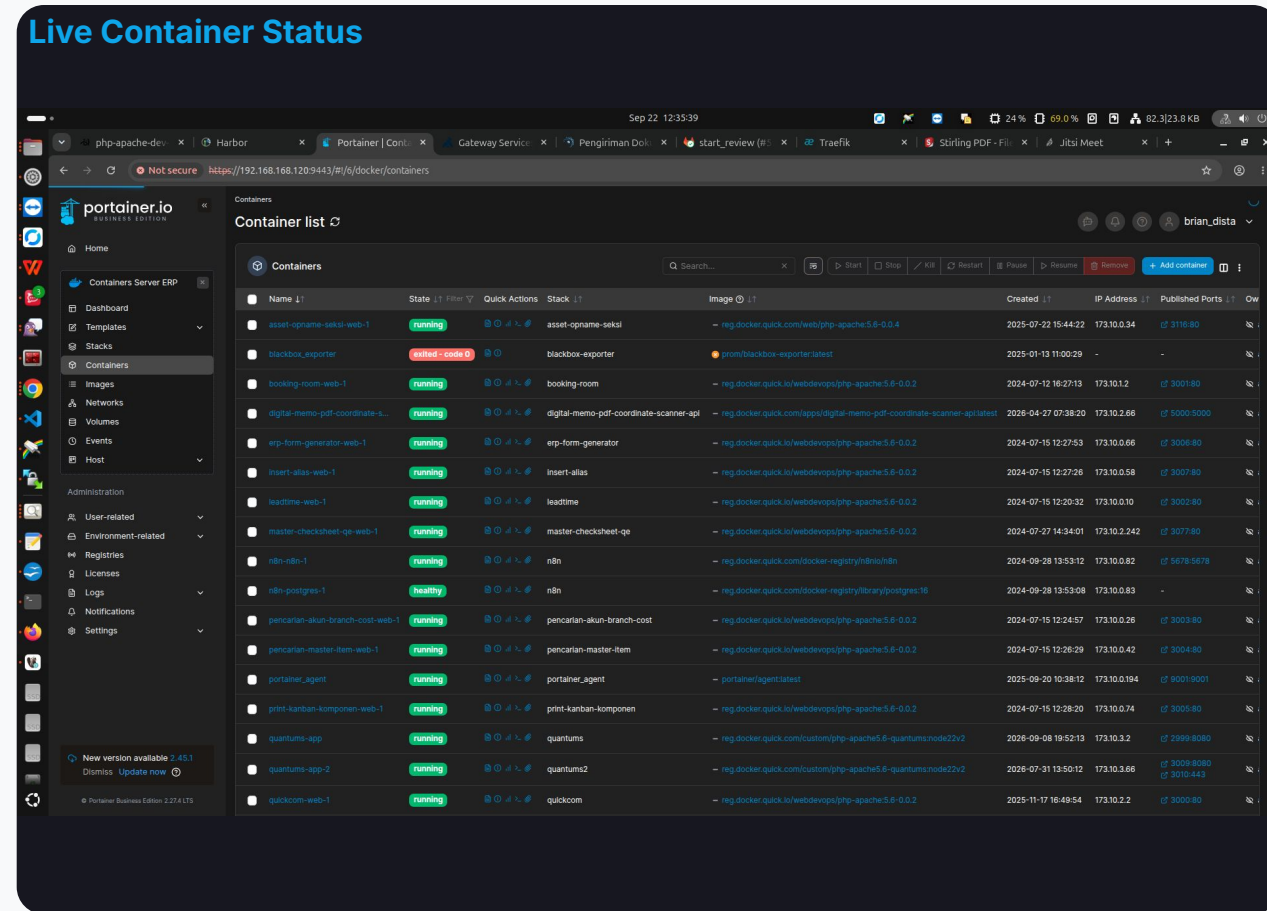
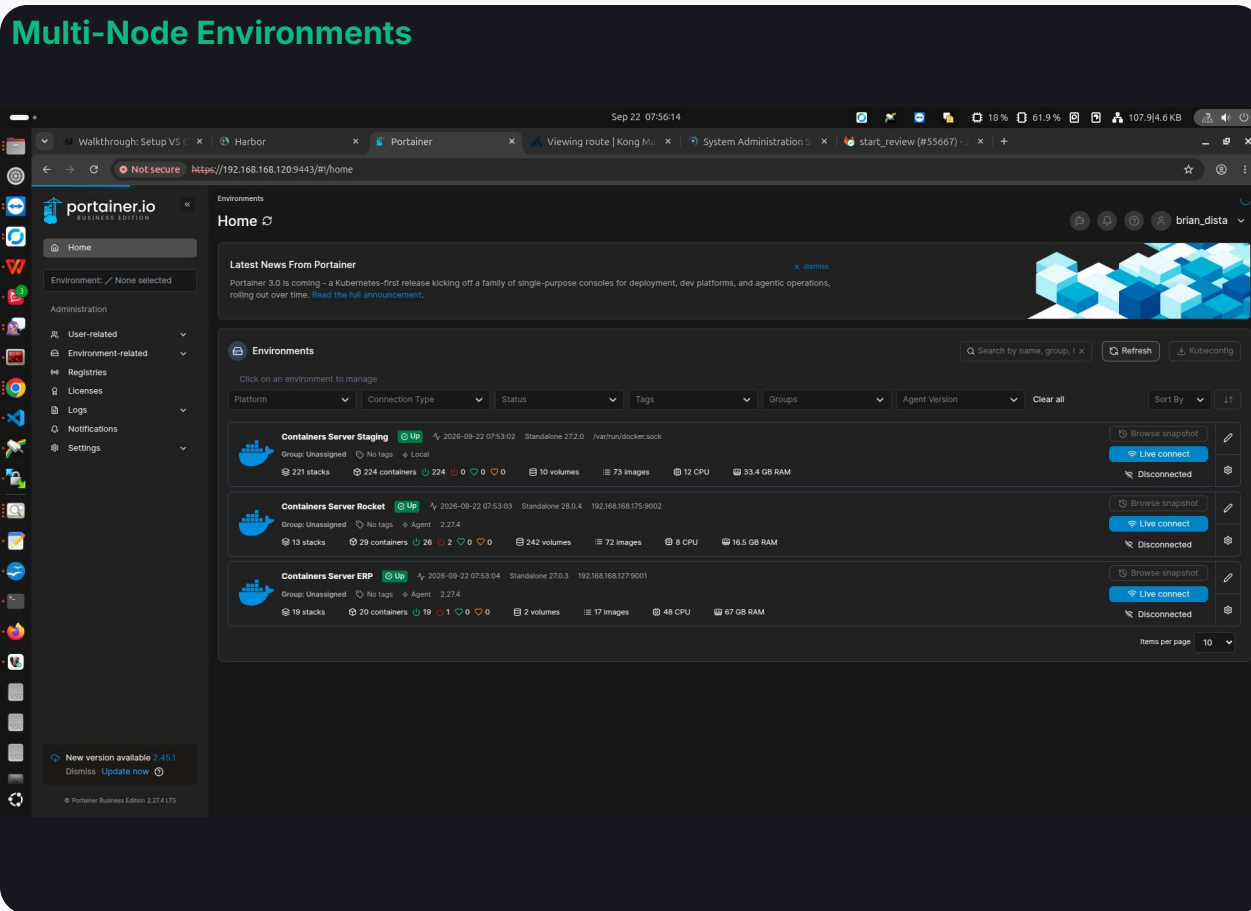
Pusat penyimpanan aman untuk mendistribusikan custom image ke seluruh environment.

Self-hosted mandiri.

Manajemen akses terkontrol untuk setiap proyek internal.

Sentralisasi Orkestrasi Portainer

Monitoring dan manajemen seluruh node dan kontainer dari satu dashboard.



Implementation By BRIAN DISTA

Strategi Deployment

Target Spesifik

Penggunaan tags pipeline untuk menunjuk server produksi yang tepat.

Resolusi Konflik

Reset sinkronisasi dan pull otomatis dari branch utama.

Manajemen Paket

Eksekusi Composer instalasi on-the-fly jika terdeteksi perubahan.



Sistem Siap Beroperasi

Lingkungan pengembangan yang lebih efisien, terisolasi, dan terotomatisasi penuh.

